

Standard Form & Indices

Test 10 · 41 marks total · Sections F & G include Higher-tier content

Name: _____

Date: _____

Score: _____ / 41

Section A — Laws of indices

A1. Write $2^8 \times 2^7$ as a single power of 2.

[1 mark]

A2. Write $3^{20} \div 3^{11}$ as a single power of 3.

[1 mark]

A3. Write $(10^4)^3$ as a single power of 10.

[1 mark]

A4. Work out the value of $6^1 \times 6^2$.

[2 marks]

Section B — Zero and negative indices

B1. Work out the value of 999^0 .

[1 mark]

B2. Work out the value of 11^{-2} . Give your answer as a fraction.

[2 marks]

B3. Work out the value of 2^{-3} . Give your answer as a decimal.

[2 marks]

Section C — Writing standard form

C1. Write 57,000 in standard form.

[2 marks]

C2. Write 6,800,000 in standard form.

[2 marks]

C3. Write 0.0093 in standard form.

[2 marks]

C4. Write 0.0000005 in standard form.

[2 marks]

Section D — Converting back, and calculating

D1. Write 4.7×10^6 as an ordinary number.

[1 mark]

D2. Write 6×10^{-5} as an ordinary number.

[1 mark]

D3. Work out $(3 \times 10^6) \times (4 \times 10^3)$. Give your answer in standard form.

[3 marks]

D4. Work out $(3 \times 10^5) \div (6 \times 10^2)$. Give your answer in standard form.

[3 marks]

Section E — Powers to know, and estimating roots

E1. Write down the value of 4^4 .

[1 mark]

E2. $\sqrt{85}$ lies between which two consecutive whole numbers?

[1 mark]

Section F — Laws of indices with algebra

F1. Simplify $n^{10} \times n^6$.

[1 mark]

F2. Simplify $r^{20} \div r^8$.

[1 mark]

F3. Simplify $(s^5)^3$.

[1 mark]

F4. Simplify $8a^2 \times 4a^7$.

[2 marks]

F5. Simplify $(5b^2)^3$.

[2 marks]

Section G — Fractional indices (Higher)

G1. Work out the value of $225^{1/2}$.

[1 mark]

G2. Work out the value of $8000^{1/3}$.

[1 mark]

G3. Work out the value of $16^{3/2}$.

[2 marks]

G4. Work out the value of $64^{-2/3}$.

[2 marks]